Q.20 Find derivative of $\dfrac{2x-4}{x^2+6}$ w.r.t. x.

Q.21 Differentiate log cos 5 x w.r.t. x.

Q.22 Evaluate tan 15°.

Section-D

Note: Long answer type questions. Attempt any two questions out of three questions. (2x8=16)

Q.23 a) Give any four methods of water proofing.

b) Explain primary colours and colour mixing.

Q.24 a) Explain basic principle of working of airconditioners and refrigerators.

b) Define reverberation? State any two methods to control reverberation.

Q.25 Evaluate $\int x^2 \cos x \, dx$

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1st Year / Architecture Engg.
Subject : Applied Science and Mathematics

Time : 3 Hrs. M.M. : 60

Section-A

Note: Multiple Choice questions. All questions are compulsory. (6x1=6)

Q.1 What does the first law of thermodynamics state?

(a) Energy can be created or destroyed

(b) Entropy always increases in universe

(c) Energy can neither be created nor be destroyed, it can convert from one form to another.

(d) Heat flows from cold body to hot body.

Q.2 The forces of attraction between molecules of same material are called

(a) Intermolecular forces

(b) Adhesive forces

(c) Cohesive forces (d) None of the above

Q.3 The formula used to calculate reverberation time is given by:

(a) Sabine's formula (b) Newton's formula

(c) Faraday's law (d) Coulomb's law

Q.4 $\tan 30° =$

(a) 1 (b) $\sqrt{3}$

(c) $1/\sqrt{3}$ (d) 0

Q.5 Sec(180°-x)=

(a) Sec x (b) -sec x

(c) Cosec x (d) -cosec x

Q.6 $\int_0^2 1dx =$

(a) 0 (b) 1

(c) 2 (d) 1/2

Section-B

Note: Objective/Completion type questions. All questions are compulsory. (6x1=6)

Q.7 Absorption coefficient is defined as ratio of ________.

Q.8 SI unit of Luminous intensity is _______

Q.9 Define photovoltaic effect.

Q.10 $\frac{d}{dx}(\sec x) =$

Q.11 $\lim_{x \to 0} \frac{\sin x}{x} =$

Q.12 $\int \sec x \tan x \, dx =$

Section-C

Note: Short answer type questions. Attempt any eight questions out of ten questions. (8x4=32)

Q.13 Explain principle and working of solar cell.

Q.14 Explain greenhouse effect.

Q.15 Explain cohesive and adhesive forces in the line set.

Q.16 State and explain principle of heat engine.

Q.17 Prove that $\frac{\sin 3A + \sin 5A}{\cos 3A + \cos 5A} = \tan 4A$

Q.18 Prove that $\cos 20 \cos 40 \cos 80 = \frac{1}{8}$

Q.19 From a point 10 meter away from the foot of a tower, the angle of elevation of the top of the tower is 45°. Find the height of the tower.